

STACK QNO:3

QUESTION:

Problem Description:

You are given an array A of Q integers and Q queries. In each query, you are given an integer i ($1 \leq i \leq N$).

Your task is to find the minimum index greater than i ($1 \leq i \leq N$) such that:

1. Sum of digits of A_i is greater than the sum of digits of A_j
2. $A_i < A_j$

If there is no answer, then print -1.

Constraints

$$1 \leq N, Q \leq 10^5$$

$$1 \leq A_i \leq 10^9$$

$$1 \leq Q_i \leq N$$

Input format

- The first line contains two numbers N and Q .
- The next line contains N numbers.
- Next Q lines contain Q queries.

Output format

Print the answer as described in the problem

▼ Logical Test Cases

Test Case 1	Test Case 2
INPUT (STDIN) 5 5 62 70 28 62 92 1 5 3 4 2 EXPECTED OUTPUT 2 -1 4 -1 -1	INPUT (STDIN) 6 6 12 71 21 16 91 11 15 31 14 21 EXPECTED OUTPUT -1 -1 -1 -1 -1 -1

▼ Mandatory Test Cases

Test Case 1	Test Case 2
KEYWORD if(arr[x]<arr[y])	KEYWORD if(arr2[x]>arr2[y])

▼ Complexity Test Cases

Test Case 1	Test Case 2	Test Case 3
CYCLOMATIC COMPLEXITY 10	TOKEN COUNT 310	NLOC 57

Source Code:

```
#include <stdio.h>
int arr[100005];
int arr2[100005];
int getDigitSum(int num) {
    int sum = 0;
    while (num > 0) {
        sum += num % 10;
        num /= 10;
    }
    return sum;
}

int main() {
    int n, q;
    if (scanf("%d %d", &n, &q) != 2) {
        return 0;
    }

    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
        arr2[i] = getDigitSum(arr[i]);
    }
```

```
    for (int k = 0; k < q; k++) {
        int query;
        scanf("%d", &query);
        int x = query - 1;
        int ans = -1;
        for (int y = x + 1; y < n; y++) {
            if(arr[x]<arr[y]) {
                if(arr2[x]>arr2[y]) {
                    ans = y + 1;
                    break;
                }
            }
        }
        printf("%d", ans);
        if (k < q - 1) {
            printf(" ");
        }
    }
    printf("\n");
    return 0;
}
```

Output:

```
5 5
62 70 28 62 92
1
2
5
-1
3
4
4
-1
2
-1
```

```
6 6
12 71 21 16 91
11
15
-1
31
-1
14
-1
21
-1
11
-1 15
-1
```